

HACKATHON COMPETITION ORAL PRESENTATION & DEFENSE MANUAL

BeatAhead: Presentation Speech & Judge Q&A

Complete Word-for-Word Presenter Scripts for Overview, Trends, and AI Insights + Winning Technical Defense

TARGET PAGES: /dashboard, /trends, /insights

MODEL FREEZE: XGBoost Matrix A ($\tau = 0.156742$)

PLATFORM: prevention-of-heart-attack-txdb.vercel.app

LIVE WEB APP

prevention-of-heart-attack-txdb.vercel.app

TOTAL PRESENTATION TIME

3 to 5 Minutes

CORE METRIC

Ischemic Stress Index (ISI, 0–100)

DATASET PROVENANCE

VitalDB (100 Surgical Cohort)

0.8896

DEVELOPMENT AUROC

~75×

PR-AUC OVER BASELINE (0.3247)

99.71%

LOCKED TEST SPECIFICITY

5 Minutes

ADVANCE EARLY-WARNING GAP

PART 1 The Overview (/dashboard) Speech & Defense

Presenter Instructions: Start with the dashboard loaded on screen. Keep the circular ISI Gauge and the Top Stat Cards in clear view. Speak with authority, energy, and clinical clarity.

[POINT TO DASHBOARD HEADER] "Judges, welcome to **BeatAhead**. Perioperative myocardial ischemia is the silent killer of patients undergoing major surgery—over 200 million patients every year. Today, standard operating room monitors only alert clinicians after the ST segment has already crashed past negative 1.0 millimeters. That is too late—tissue damage has already started.

[POINT TO ISIGAUGE (CIRCULAR 0-100 METER)] Here on our **Overview Dashboard**, you see the core breakthrough: our **Ischemic Stress Index (ISI)**. Rather than showing an isolated heart rate or raw ECG wave, the ISI synthesizes continuous electrophysiology, autonomic strain, vascular perfusion, and our validated machine learning model into a unified, actionable clinical index bounded strictly between 0 and 100.

[POINT TO TOP STATS CARD & MODEL STATUS BADGE] In our live stats card, you can see our frozen Phase 5 XGBoost research model evaluating real-time physiological vectors. Notice the operating decision threshold of $\tau = 0.156742$. Our model probability is sitting at 0.005, safely in the 'Normal / Stable' baseline tier.

[POINT TO BASELINECARD & CONTRIBUTIONBARS] Crucially, look at the bottom two cards: the **Personal Baseline Tracker** and the **Physiological Contribution Breakdown**. BeatAhead does not rely on generic population cutoffs. It continuously models each patient's resting baseline using a Gated Exponential Moving Average, guaranteeing that normal variations in resting heart rate or arterial tone never cause false alarms."

Expected Judge Questions & Winning Answers for Overview (/dashboard)

Q1 What exactly does this 0 to 100 ISI score mean, and why is it better than conventional hospital monitors?

Answer: "Conventional monitors are single-parameter and reactive: they beep only when ST segment depression breaches ≤ -1.0 mm or heart rate crosses 120 bpm. The Ischemic Stress Index (ISI) is a multi-modal composite index (0–100) that measures *impending subendocardial oxygen supply-demand deficit*. It combines:

- **Model Evidence ($W_{\text{model}} = 0.45$):** The output of our frozen XGBoost model.
- **Autonomic Strain ($W_{\text{auto}} = 0.30$):** Sympathetic activation measured by sudden heart rate surges and RMSSD parasympathetic withdrawal.
- **Vascular Perfusion Deficit ($W_{\text{perf}} = 0.25$):** Pulse Arrival Time (PAT) transit elongation and SpO_2 desaturations. It transforms fragmented monitor beeps into a single predictive score with a 5-minute advance early-warning horizon."

Q2 How does your personal baseline work, and what prevents a deteriorating patient's baseline from 'adapting' to the disease?

Answer: "We engineered a proprietary **Gated Anti-Drift EMA Filter**. During the initial 10 minutes of stable monitoring, the system computes the patient's individual resting baseline for heart rate, SDNN, and PAT transit time. As long as physiology remains stable, the baseline updates smoothly ($\alpha = 0.02$).

However, the instant the ISI score elevates or the ML model detects pre-ischemic drift, **baseline adaptation locks completely**. This prevents the physiological baseline from 'chasing' the pathological state and normalizing dangerous ischemia."

Q3 What is this odd threshold number $\tau = 0.156742$ shown in the stats card?

Answer: "In surgical telemetry, true sustained ischemia is an extreme minority event—natural clinical prevalence in our 80-case development cohort was just **0.43%** (37 positive events in 8,571 windows). A naive 0.50 threshold would predict zero alerts and miss every patient.

We calibrated $\tau = 0.156742$ mathematically on pooled out-of-fold predictions to maximize the F_1 -score ($\arg\max(F_1)$). At this frozen threshold, BeatAhead achieved an outstanding **99.71% specificity** and a tiny false alarm rate of **0.173 alarms per surgical hour**—less than 1 false alarm every 5.77 hours."

PART 2 The Trends (/trends) Speech & Defense

Presenter Instructions: Click the **Trends** link in the left sidebar. As the page loads, point to the 24h/7d/30d filter tabs and the multi-signal stacked graphs.

[CLICK /TRENDS & POINT TO TIME FILTERS] "Now, let's step into **Long-term Trends**. Acute cardiac events do not happen in a vacuum—they are preceded by subtle circadian shifts, progressive autonomic fatigue, and micro-instabilities over hours and days.

[POINT TO STACKED LINE CHARTS: ISI, HRV, SPO2, HR, MOTION] Here, clinicians can inspect synchronous historical patterns across 24-hour, 7-day, and 30-day horizons. Notice how the graphs align:

- Top Red: The composite **ISI Trend** over time.
- Dark Slate: **HRV (SDNN)**, demonstrating whether the parasympathetic nervous system is recovering or chronically suppressed.
- Blue: **\$SpO_2\$ Stability**, capturing nocturnal desaturations and micro-hypoxia.
- Purple: **Activity & Motion Intensity** from the patient's wearable or surgical IMU.

[SCROLL DOWN TO INSIGHT TIMELINE] At the bottom is our **Insight Timeline**. Rather than forcing a clinician to scroll through millions of raw data points, our pipeline clusters events into color-coded physiological states: normal circadian rhythm, physical exertion, ischemic stress surges, and post-event physiological recovery. It gives doctors a 10-second longitudinal summary of the patient's cardiovascular trajectory."

Expected Judge Questions & Winning Answers for Trends (/trends)

Q4 How do longitudinal trends help prevent cardiac events compared to real-time alerts?

Answer: "Real-time alerts save lives during an acute crisis; longitudinal trends prevent the crisis from developing in the first place. For example, if a patient exhibits an upward-trending nocturnal ISI slope over 3 consecutive nights—coupled with progressive HRV suppression and intermittent \$SpO_2\$ dips—it signals microvascular decompensation, silent ischemia, or worsening heart failure. A cardiologist can adjust beta-blockers, nitrates, or ACE inhibitors days before the patient ever suffers a catastrophic acute myocardial infarction."

Q5 How does your trend engine handle sensor disconnections or missing data?

Answer: "Our Signal Quality Index (SQI) filters continuous streams. When the sensor is removed or signal quality drops below 0.35: 1. The window is tagged as '**Insufficient Signal Quality**' or '**artifact**' rather than imputing false zeros or hallucinatory numbers. 2. On the Insight Timeline, you can see these explicitly marked with amber and red badges. 3. In trend aggregation, our median-imputer ignores corrupted intervals so that long-term averages remain clinically pristine."

Q6 Why do you track Activity/Motion alongside cardiac vitals in the trends view?

Answer: "Because physiological context is everything. An elevated heart rate of 115 bpm with high motion intensity indicates normal, healthy physical exercise. However, a heart rate of 115 bpm with zero motion intensity and suppressed HRV represents acute sympathetic overdrive or ischemic distress. Tracking motion intensity allows our algorithm to contextualize tachycardia and eliminate false positive exercise alerts."

PART 3 The AI Insights (/insights) Speech & Defense

Presenter Instructions: Click **AI Insights** in the sidebar. This is where you blow the judges away with machine learning rigor, explainability, and scientific validation.

[CLICK /INSIGHTS & POINT TO TOP RESEARCH MODEL CARD] "Finally, let's look at **AI Insights**—the transparency and explainability engine of BeatAhead. In critical healthcare, black-box AI is unacceptable. Doctors will not trust a system that simply beeps without explaining why.

[POINT TO RESEARCH MODEL SIGNAL CARD] At the top, you see our **Frozen XGBoost Research Model Card**. This model was trained on 100 multi-modal surgical cases from the VitalDB database using an audited 26-feature physiological schema across cardiac repolarization, pulse transit, and oxygen saturation.

[POINT TO CONTRIBUTIONBARS & COMPONENT SUMMARY TABLE] Below the model card, BeatAhead deconstructs the prediction into exact physiological contributors:

- **Model Evidence (\$E_{model}\$):** Transformed via a piecewise exponential function centered on threshold 0.156742.
- **Autonomic Strain (\$D_{auto}\$):** Quantifying vagal withdrawal and sympathetic surge.
- **Vascular Perfusion (\$D_{perf}\$):** Tracking Pulse Arrival Time elongation from the R-peak to the PPG foot delay.
- **Motion & Quality Gating:** Displaying the exact confidence impact of movement artifacts.

[HIGHLIGHT REGULATORY & EXPLAINABILITY NOTE] By combining TreeSHAP feature importance with physiological component attribution, BeatAhead provides clinicians with immediate, interpretable evidence, transforming an artificial intelligence prediction into a trusted clinical decision support tool."

Expected Judge Questions & Winning Answers for AI Insights (/insights)

Q7 Why did you use XGBoost instead of a Deep Learning Transformer or LSTM for the AI model?

Answer: "We benchmarked gradient-boosted trees against deep neural architectures and chose XGBoost for three decisive clinical reasons:

1. **Explainability:** XGBoost enables exact TreeSHAP attribution in sub-milliseconds, allowing us to display the physiological contribution bars you see right here.
2. **Sample Efficiency on Extreme Class Imbalance:** On 100 surgical cases with 37 true ischemic events, deep neural networks overfit and hallucinate. Regularized shallow gradient-boosted trees (max depth 4, scale_pos_weight 230.65) achieved superior out-of-fold generalization (0.8896 AUROC).
3. **Edge Latency:** Inference executes in **under 2 milliseconds** on a standard CPU, enabling instant client/serverless execution without expensive GPU cloud latency."

Q8 What was your feature ablation study, and what did it prove about wearable signals?

Answer: "We conducted an ablation benchmark between **Matrix A** (all 26 features including ST segment repolarization) and **Matrix B** (purely non-ST vascular and autonomic features like PAT, HRV, perfusion index, and \$SpO_2\$).

Matrix B collapsed to an AUROC of **0.53** (essentially random). This provided scientific proof that *wearable vascular signals alone cannot predict myocardial ischemia in advance without cardiac electrophysiological repolarization features*. BeatAhead requires multi-modal synthesis."

Q9 Explain the math behind Model Evidence (\$E_{model}\$). Why not just use \$P_{model}\$ times 100\$?

Answer: "Because our operating threshold is $\tau = 0.156742$ due to extreme class imbalance, a raw probability of 0.15 represents high clinical risk! If we simply multiplied 0.15×100 , the patient would see an ISI of 15, which looks completely safe.

We solved this by designing a **piecewise exponential transfer function**:

- Below τ : Conservative sub-linear expansion $(P/\tau)^{1.4} \times 0.50$, anchoring baseline risk.
- Above τ : Rapid clinical alert ramp $0.50 + 0.50 \times ((P - \tau)/(1 - \tau))^{0.25}$, rapidly elevating the ISI into the elevated risk tier the moment τ is crossed."

PART 4 The Unified 3-Minute Judge Demo Master Script

[0:00 - 0:45 | Dashboard Hook & Problem]

"Judges, every year 200 million patients undergo major surgery, and intraoperative myocardial ischemia is the leading cause of surgical mortality. Hospital monitors today are purely reactive: they only beep after the ST segment collapses below -1.0 mm. We built **BeatAhead** to give surgical teams a **5-minute advance early warning** with 99.71% specificity. Here on the **Dashboard Overview**, our Ischemic Stress Index (ISI) synthesizes multi-parameter vitals into an intuitive 0–100 score, calibrated against each patient's personal resting baseline."

[0:45 - 1:30 | Long-Term Trends & Context]

[\[CLICK /TRENDS\]](#) "Moving to **Long-Term Trends**, BeatAhead tracks longitudinal cardiac patterns across 24-hour, 7-day, and 30-day windows. By correlating the ISI score against parasympathetic HRV recovery, nocturnal \$SpO_2\$ stability, and motion intensity, clinicians can spot progressive microvascular decompensation days before an acute infarction occurs."

[1:30 - 2:30 | AI Insights & Scientific Rigor]

[\[CLICK /INSIGHTS\]](#) "In high-stakes medicine, black-box AI is unacceptable. In **AI Insights**, you can see our frozen Phase 5 XGBoost model trained on 100 VitalDB surgical patients. It achieves an **0.8896 AUROC** and an average precision **75 times higher than chance**, with a tiny false alarm rate of 0.17 alarms per hour. Every alert is mathematically deconstructed into Model Evidence, Autonomic Strain, and Vascular Transit Time, giving clinicians complete transparency."

[2:30 - 3:00 | Impact & Closing]

"BeatAhead is fully deployed and production-ready on Vercel at prevention-of-heart-attack-txdb.vercel.app. By turning retrospective surgical monitors into predictive, explainable clinical decision support, BeatAhead helps save lives and saves hospitals over \$750,000 annually. Thank you, and we welcome your questions!"

PART 5 Quick-Reference Numbers to Memorize for Q&A

Metric	Value	What to Say to Judges
Development AUROC	0.8896	Discriminates pre-ischemic micro-drifts from normal surgical physiology.
Precision-Recall AUC	0.3247	~75 times higher than the natural clinical prevalence baseline of 0.0043.
Locked Test Specificity	99.71%	Guarantees clinicians are not bombarded with noisy alarms.
False Alarm Rate	0.173 / hour	Fewer than 1 false alert every 5.77 surgical hours—solves alarm fatigue.
Advance Lead Gap	5 Minutes	Enforced 300s blank gap ensures true predictive warning with zero data leakage.
Operating Threshold	$\tau = 0.156742$	Calibrated via out-of-fold F1 optimization to tackle extreme class imbalance.
Inference Latency	< 2 milliseconds	Ultra-fast CPU evaluation; runs seamlessly on serverless edge nodes.